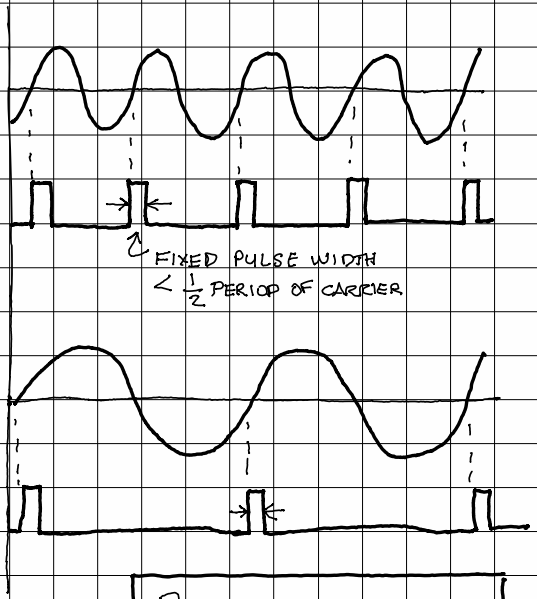
PULSE COUNTING FM DISCRIMINATOR

W2AEW

- FM DEMODULATOR THAT DOESN'T RELY ON TUNED CIRCUITS LIKE THE TRADITIONAL FM RATIO DETECTOR OR FOSTER-SEELEY DISCRIMINATOR

- **HOW IT WORKS:**
  - GENERATE NARROW FIXED-WIDTH PULSES AT FM CARRIER (IF) FREQUENCY (AT ZERO-CROSSING, FOR EXAMPLE)
  - THE DUTY-CYCLE VARIES WITH THE CARRIER (IF) FREQUENCY
  - INTEGRATE / AVERAGE THE PULSES TO RECREATE THE BASEBAND (AUDIO)
  - WHEN f_c IS LOW...
    ... FEWER PULSES, AVERAGE VOLTAGE IS LOW
  - WHEN f_c INCREASES...
    ... MORE PULSES, AVERAGE VOLTAGE INCREASES



PULSES CAN BE POSITIVE OR NEGATIVE

PULSE COUNTING FM DEMODULATOR - MORE DETAILS

- ALMOST ALWAYS REQUIRES DOWN-CONVERSION FROM ORIGINAL CARRIER FREQUENCY INTO A LOWER FREQUENCY IF (INTERMEDIATE FREQ)

WHY?

- PEAK-TO-PEAK FREQUENCY DEVIATION IS TYPICALLY MUCH, MUCH, MUCH SMALLER THAN CARRIER FREQUENCY

∴ THE DUTY-CYCLE VARIATION OF THE PULSE TRAIN IS VERY SMALL

EXAMPLE: $f_c = 100\,MHz$ $f_{DEV} = 50\,KHz$
(FREQ VARIES FROM 99.95MHz TO 100.05MHz)

- WHEN YOU DOWN-CONVERT...
  - CARRIER FREQUENCY IS REDUCED
  - DEVIATION REMAINS THE SAME.

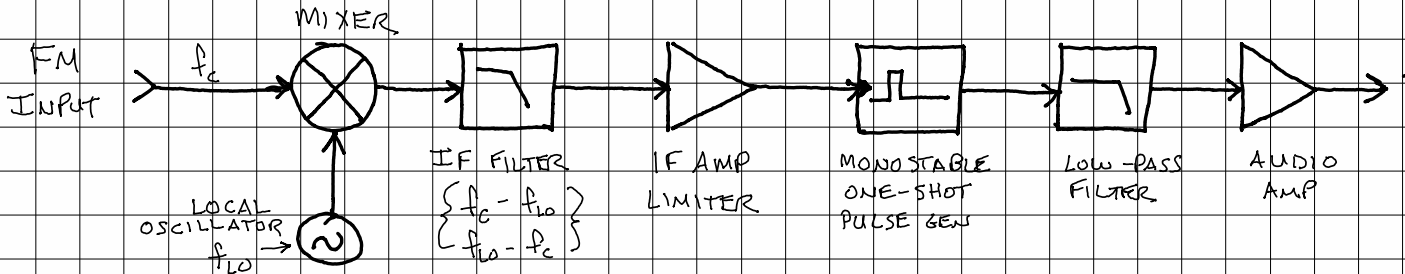
∴ THE P-P FREQUENCY DEVIATION IS A LARGE PORTION OF IF FREQ.

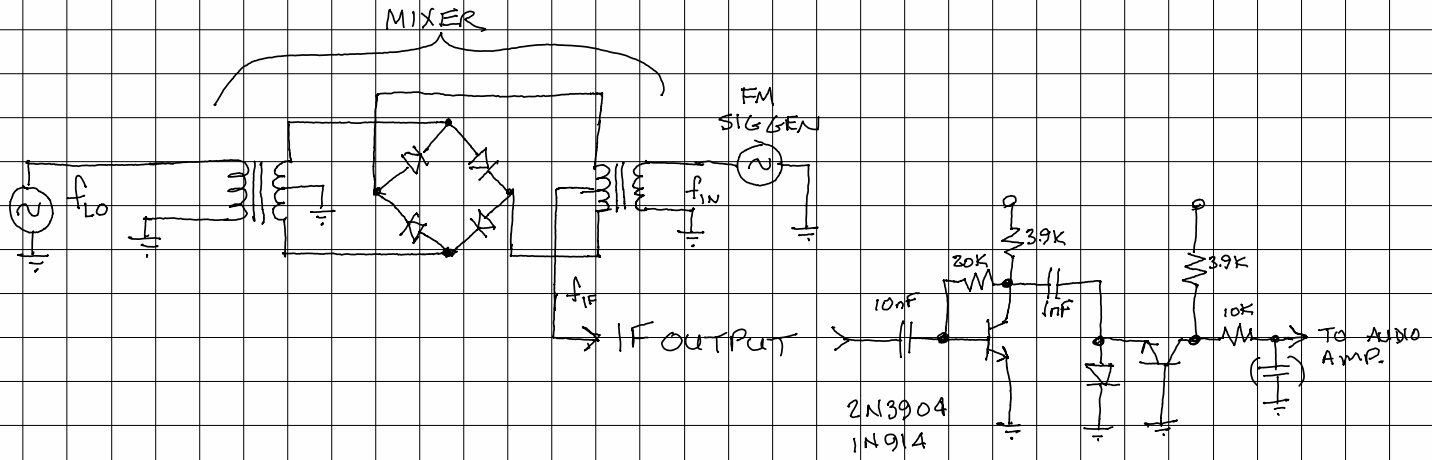
EXAMPLE: $f_{IF} = 100\,KHz$ $f_{DEV} = 50\,KHz$
(FREQ VARIES FROM 50KHz TO 150KHz)

PRACTICAL TIPS

- $f_{IF} >$ PEAK DEVIATION
- LOWER $f_{IF} \rightarrow$ LARGER BASEBAND O/P
- HIGHER $f_{IF} \rightarrow$ EASIER TO FILTER

CIRCUITS


FM INPUT
f_c
MIXER
LOCAL OSCILLATOR
f_{LO}
IF FILTER
$\{f_c - f_{LO}$
$f_{LO} - f_c\}$
IF AMP LIMITER
MONOSTABLE ONE-SHOT PULSE GEN
LOW-PASS FILTER
AUDIO AMP


MIXER
f_{LO}
FM SIGGEN
f_{IN}
f_{IF}
IF OUTPUT
10nF
20K
3.9K
1nF
3.9K
10K
TO AUDIO AMP.
2N3904
1N914